

# SHAIK FAYAZ BASHA

AI/ML Engineer · Python Developer

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## PROFESSIONAL SUMMARY

AI/ML Engineer with 1+ year of hands-on experience developing and deploying Machine Learning, Deep Learning, Generative AI, Computer Vision, NLP, and Retrieval-Augmented Generation (RAG) solutions. Proficient in Python, TensorFlow, PyTorch, Scikit-Learn, OpenCV, LangChain, and OpenAI API for building intelligent, data-driven applications. Skilled in model training, transfer learning, fine-tuning, prompt engineering, and end-to-end AI application development using Django, FastAPI, REST APIs, and vector databases. Proven experience delivering production-ready AI solutions that solve real-world business and research challenges.

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, C, JavaScript, HTML, CSS

**AI / ML Frameworks:** TensorFlow, PyTorch, Keras, Scikit-Learn, Hugging Face Transformers, LangChain, NLTK, OpenCV

**Generative AI & LLMs:** OpenAI API, LLMs, RAG Pipelines, Prompt Engineering, Agentic AI, Fine-Tuning, Transfer Learning, Sentence Transformers

**Computer Vision & NLP:** YOLOv8, ResNet, LeNet, CNN, Object Detection, Image Classification, Audio Classification, Mel-Spectrogram, Speech Recognition

**Vector Search & Databases:** FAISS, ChromaDB, Vector Databases, Semantic Search, Embedding Models

**Cloud & Tools:** Firebase, Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Kaggle

**Web & Databases:** Django, FastAPI, SQLite, MySQL, Backend Development

**ML Workflow:** Feature Engineering, Feature Selection, Hyperparameter Tuning, Model Evaluation, ETL, GPU Training

## WORK EXPERIENCE

**Independent Projects & Research** — Chennai, Tamil Nadu

Aug 2025 – Present

### AI/ML Engineer

- Built and deployed 5+ end-to-end AI/ML applications using Python, TensorFlow, PyTorch, and Django across Computer Vision, NLP, and Generative AI domains.
- Developed RAG chatbot using LangChain, FAISS vector indexing, and OpenAI API — achieved 90%+ contextual answer relevance across 5,000+ document chunks with sub-500ms latency.
- Trained YOLOv8 and ResNet models for real-time object detection; achieved 91% mAP at 30+ FPS, deployed via FastAPI REST endpoints with Docker containerization.
- Applied Transfer Learning and Fine-Tuning on pre-trained models (ResNet, BERT), reducing training time by 45% while maintaining 90%+ accuracy on custom datasets.
- Implemented CI/CD pipelines using GitHub Actions; containerized 3 AI applications with Docker, reducing deployment time by 60%.
- Built multimodal emotion detection system — CNN image (94% accuracy), NLP Transformer text (92% accuracy), audio CNN (90% accuracy) — with OpenAI-powered therapy engine and SMTP alert system.
- Integrated Firebase real-time database and REST APIs into AI monitoring dashboard; served 100+ concurrent users via Django and FastAPI backends.

**Supreme Info Solutions** — Nellore, Andhra Pradesh

Jan 2025 – Apr 2025

### Machine Learning Intern

- Built Emotion-Based Music Recommendation System using OpenCV + CNN (7 emotion classes); delivered real-time emotion-to-music mapping in under 500ms via Django web application.
- Executed full ML lifecycle: data preprocessing, EDA, feature engineering, model training, hyperparameter tuning, and model deployment in a production environment.

EDUCATION

Master of Computer Applications (MCA)  
Narayana Engineering College, Nellore — JNTUA University

Aug 2023 – Apr 2025

Bachelor of Science – Computer Science (BSc)  
Sri Venkateswara University, Tirupati

Jun 2020 – Apr 2023

FEATURED PROJECTS

1. Ayurvedic Knowledge Assistant – RAG-Based Medical Information Chatbot

Tech Stack: Python, LangChain, RAG, LLM, Hugging Face Transformers, FAISS, ChromaDB, Sentence Transformers, PDF Processing, NLP, Django, REST API, SQLite

- Built an AI-powered RAG chatbot for Ayurvedic healthcare knowledge using LangChain + FAISS/ChromaDB, enabling natural language queries across 300+ pages of Ayurvedic PDF documents with 90%+ contextual answer relevance.
- Engineered end-to-end document intelligence pipeline — PDF extraction, text preprocessing, semantic chunking, and Sentence Transformer embedding generation — achieving sub-500ms semantic retrieval from 10,000+ indexed document chunks.
- Integrated LLMs via LangChain retrieval chains with Prompt Engineering (chain-of-thought, few-shot), reducing hallucination rate by 35% through retrieval-based augmentation and source-grounded response generation.
- Deployed full-stack Django web application with REST API endpoints supporting real-time conversational AI, document ingestion, and scalable vector search from Ayurvedic literature.

2. Multimodal Emotion Detection System with LLM Micro-Therapy & Smart Alert Engine

Tech Stack: Python, TensorFlow, PyTorch, OpenCV, CNN, NLTK, Hugging Face Transformers, Librosa, OpenAI API, LangChain, Django, REST API, SMTP Email API, SQLite

- Engineered hybrid multimodal AI system fusing 3 pipelines — facial CNN (94% accuracy), NLP Transformer text classifier (92% accuracy), audio mel-spectrogram CNN (90% accuracy) — via majority-voting ensemble logic.
- Integrated OpenAI API + LangChain Agentic AI for real-time personalized micro-therapy suggestions; implemented SMTP email alert system for high-risk emotions (Sad/Angry/Distressed).
- Applied Transfer Learning on Hugging Face pre-trained Transformers for text emotion classification, reducing training data requirement by 60% while achieving 92% accuracy.
- Deployed complete system via FastAPI REST API + Django dashboard; real-time live video emotion detection at 25+ FPS; containerized with Docker for production deployment.

LANGUAGES

- ENGLISH
- HINDI
- TELUGU
- TAMIL
- URDU